

Documentation of Remaining Analytical Limitations

HealthConnect Platform Validation & Risk Disclosure

Project: HealthConnect Testing & Validation Framework | **Scope:** Analytical Governance & Constraints

While the HealthConnect validation and refinement process successfully resolved core data mapping issues and established reliable predictive indicators, several structural analytical limitations must be documented for future monitoring phases:

1. Historical Data Gaps

Legacy Attrition & Missing Entries

Legacy records prior to the enforcement of mandatory logging contain unrecorded entries in secondary attributes (such as `historical_reminder_channel` or `detailed_waiting_time_minutes`), which could introduce minor sampling bias in retrospective analyses.

2. Observational vs. Causal Inference

Statistical Correlation Limits

Validated findings establish strong statistical correlations (e.g., between `booking_lead_days` and appointment no-shows), but they remain observational. Controlled pilot interventions or A/B testing are still required to confirm direct causal impact.

3. Uncaptured External Confounders

External Environmental Variables

The current data model does not account for external real-world variables—such as local transit disruptions, severe weather events, or sudden economic constraints—which can independently influence patient attendance.

4. Temporal Scope & Longitudinal Shift

Static Snapshot Constraint

The dataset represents a specific operational snapshot in time. Longitudinal behavioral shifts or seasonal healthcare demand cycles may alter risk weights over extended periods, necessitating ongoing model recalibration.